Somil Doshi

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Summary

Graduate Engineering Data Science student with experience across data analytics, data engineering, and applied analytics and modeling in academic and industry settings. Skilled in building end-to-end data pipelines, performing data transformations, and delivering insights through dashboards, reporting, and scalable data workflows.

Professional Experience

University of Houston — **Mar 2025 – Present**
Research Assistant - Data Engineering and Analytics

- Designed and maintained end-to-end data pipelines for research datasets, supporting ingestion, transformation, and structured storage, improving data availability and analysis readiness by 30%.
- Built and optimized **SQL**-based analytical data models using **Databricks**, reducing query latency by 22% and improving analytical performance across research workflows.
- Deployed and managed data services and storage systems using **PostgreSQL**, **Docker**, and **Kubernetes**, ensuring scalable, secure access to datasets and supporting multiple concurrent research projects.

Oasis Infobyte — **Sept 2022 – May 2023**
Data Analyst Intern

- Consolidated 50K+ energy usage records in **MongoDB**, **Databricks** with Airflow orchestration, applying the DBT transformations to align, improve forecasting reliability by 19% across multiple downstream analytics workflows.
- Configured time-series analytics pipelines in **Databricks** and **AWS SageMaker**, integrated with **Kafka** and validated through **Airflow** DAGs, sustaining 92% backtesting accuracy and reducing prediction error margins by 18%.
- Built **Power BI** dashboards with DAX to visualize seasonal trends and generate anomaly alerts for analytics teams.

TCR Innovations — **Jan 2022 – Aug 2022**
Data Engineer Intern

- Introduced ETL pipelines in Python with **AWS RDS** integration, managing 80K+ drilling records and enforcing schema rules, reducing refresh cycles by 34% and improving reliability of downstream workloads.
- Deployed predictive APIs supporting operational analytics, linked to **React.js** interfaces, sustaining 89% accuracy while reducing manual intervention for operational teams.
- Automated cloud deployments with Docker, **AWS Lambda**, ensuring stable builds across staging and production.

Technical Skills

- **Programming Languages & Tools** - Python, SQL, R, Java, Streamlit, Flask, AWS SageMaker, Git, MS Excel
- **Data Engineering & Deployment** - Airflow, DBT, PySpark, Docker, Kubernetes, GitHub Actions, ETL Pipelines
- **Databases & Visualization** -PostgreSQL, AWS RDS, MongoDB Atlas, Azure Cosmos DB, MySQL, Power BI, Tableau
- **Data Modeling & Processing** - Data Modeling, Data Quality Checks, Batch Processing, JSON, Parquet
- **Relevant Skills** - Analytical Thinking, Communication, Team Collaboration, Project Management, Continuous Learning

Education

University of Houston — **Aug 2024 – May 2026**
Master of Science in Engineering Data Science — *GPA: 3.89/4.0*

University of Mumbai — **Aug 2020 – May 2024**
Bachelor of Engineering in Information Technology — *GPA: 3.84/4.0*

Projects

RevPAR Intelligence Platform - BroadVail Capital | *Python, Streamlit, OpenAI API, XGBoost, LightGBM, SHAP*

- Built a revenue forecasting pipeline on 12,800+ multifamily property records, achieving 70.6% R^2 and 0.109 RMSE, reducing prediction error by 18% over baseline models.
- Automated executive reporting using LLM-driven summaries and SHAP-based explainability, delivering interpretable investment insights and validating 15-minute city performance patterns via interactive dashboards.

Predictive Maintenance Analytics for Oil Wells | *Neural Networks, PostgreSQL, Docker, Streamlit*

- Built data pipeline for drilling sensor data, improving maintenance lead time by 7 days through early issue detection.
- Reduced downtime by 28% via data-driven maintenance scheduling across 5,000+ simulated well operations.

Brain Tumor MRI Classification | *3D CNNs, PyTorch, Flask APIs, MongoDB Atlas, Docker*

- Implemented a data pipeline for MRI inference, achieving 94% classification accuracy across multiple tumor subtypes.
- Logged and served predictions via Flask with MongoDB Atlas, reducing diagnostic reporting time by 30% overall.